Pokédex Tutorial

A Casual User's Guide to Building and Using the App

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About this guide: This document is for *casual users* — people who want to run the app, understand what it does, and maybe tweak a few settings. It assumes very basic familiarity with computers. If you're comfortable with the command line, also see `docs/run.md`. If you want to modify the code, start with `docs/architecture.md`.

Contents

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1 What Is This App?

The **Pokédex Tutorial** is an iOS app that acts like a Pokédex — an encyclopedia of Pokémon creatures. You can:

- **Browse** all 1,300+ Pokémon, with official artwork
- **Search** for any Pokémon by name
- **Filter** by type (Fire, Water, Grass, etc.)
- **View details**: base stats, abilities, type effectiveness
- **Save favorites** for quick access
- **Switch** between list and grid layouts

But this app is also a **teaching tool**. Every file is heavily commented to explain *why* the code is written the way it is. It is designed to teach modern iOS development with Swift and SwiftUI, from basic types to advanced networking and testing.

2 Getting Started

2.1 What You Need

- A Mac computer running **macOS 14 (Sonoma)** or newer
- **Xcode 15.2** or newer (free from the Mac App Store)
- An internet connection (for live Pokémon data)
- About 200 MB of free disk space

2.2 Opening the Project

1. Locate the `swift-pokedex-tutorial` folder on your Mac.
2. Double-click the file named `PokedexTutorial.xcodeproj`.
   (It has a blue icon with a hammer.)
3. Xcode will open. Wait a moment while it indexes the files.
4. At the top-left of Xcode, you'll see a device selector. Click it and choose any iPhone simulator (e.g., "iPhone 17 Pro").

2.3 Running the App

1. Press **Cmd + R** (hold the Command key and press R).
2. Xcode will build the app. The first build takes 30–60 seconds.
3. The Simulator opens automatically. The app appears and loads the Pokémon list from the internet.

2.4 Running Without Internet

If you don't have internet, you can use local mock data:

1. In Xcode, open `PokedexTutorial/App/PokedexTutorialApp.swift`.

2. Find the line that says:

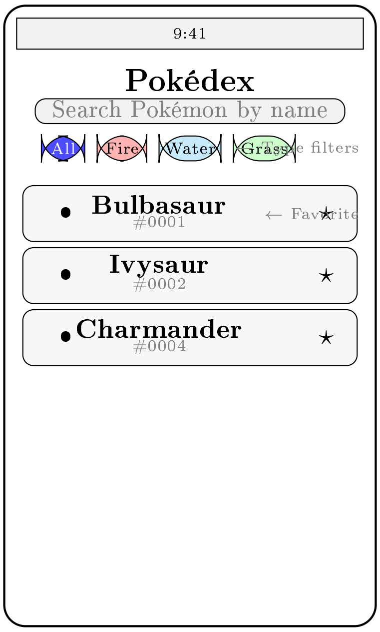
   `@StateObject private var pokemonService: PokemonAPIService = PokemonAPIService()`
3. Replace it with:

   `@StateObject private var pokemonService = MockPokemonService()`
4. Press Cmd + R to rebuild. The app now shows 10 sample Pokémon instantly.

3 Using the App

3.1 The Main Screen

When you launch the app, you see:



3.1.1 Key Interactions

Scroll down to see more Pokémon. The app loads 20 at a time (infinite scroll).

Pull down from the top to refresh the list with fresh data.

Tap the search bar at the top and type a Pokémon name. The list filters as you type.

Tap a type chip (Fire, Water, Grass, etc.) to show only Pokémon of that type. Tap "All" to clear the filter.

Tap the star ⋆ on any row to mark it as a favorite.

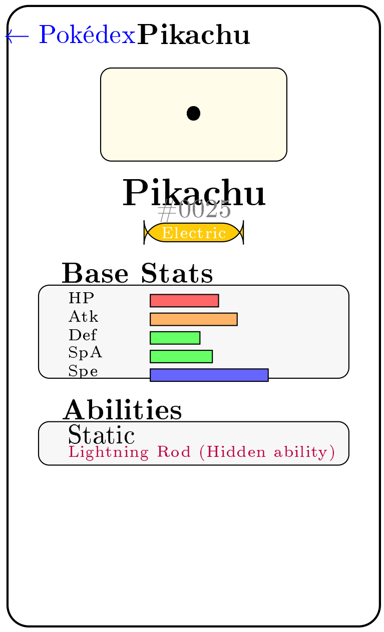
Tap the star button in the toolbar (top-right) to show only your favorites.

Tap the grid button (four squares icon) to switch between list and grid layout.

Tap any Pokémon to see its full details.

3.2 The Detail Screen

When you tap a Pokémon, you see:



3.2.1 Detail Screen Features

- **Official artwork** at the top, on a tinted background matching the Pokémon's type
- **Base stats** with color-coded bars: red (low), orange (medium), green (high), blue (exceptional)
- **Abilities** listed below stats; hidden abilities are marked in purple
- **Type Effectiveness** (expandable section, tap to open) shows what types the Pokémon is weak to, resistant to, and immune to
- **Star button** in the toolbar to favorite/unfavorite

4 Favorites Explained

Favorites are saved **even after you close the app**. They are stored on your Mac using a technology called UserDefaults.

- **Add a favorite:** Tap the ⋆ (star) icon on any Pokémon row. It turns yellow.
- **Remove a favorite:** Tap the yellow ⋆ again. It turns gray.
- **View only favorites:** Tap the star button in the toolbar (top-right corner). Tap again to show all.
- **Favorites survive restarts:** Close and reopen the app — your favorites are still there.

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5 Troubleshooting

5.1 The app shows a loading spinner forever

This usually means the PokéAPI (free service we use for data) is unavailable or your internet is down.

1. Check your internet connection.
2. Wait a minute — the PokéAPI has rate limits. Too many rapid requests will be blocked temporarily.
3. Switch to mock data: edit `PokedexTutorialApp.swift` to use `MockPokemonService()` instead of `PokemonAPIService()`.

5.2 The app shows "No Pokémon Found"

You may have an active search or filter that matches nothing.

1. Clear the search bar (tap the **X** button or delete all text).
2. Make sure the "All" type chip is selected (it should be filled, not outlined).
3. If the star filter is active (yellow star in toolbar), tap it to show all Pokémon.

5.3 Xcode says "Build Failed"

1. Make sure you're opening `PokedexTutorial.xcodeproj`, not individual `.swift` files.
2. In Xcode, go to **Product** → **Clean Build Folder** (hold Option key to see this), then try building again.
3. Make sure your selected simulator is running iOS 17.0 or newer.
4. Check that Xcode is version 15.2 or newer (Apple menu → About Xcode).

5.4 Tests fail with network errors

The API tests need an internet connection. If they fail:

1. Check your internet connection.
2. Run them again — the PokéAPI may have been temporarily rate-limiting.
3. All other tests (model tests, ViewModel tests) pass without internet.

6 Learning Path

This app is designed to teach Swift and iOS development. Here's the recommended reading order:

1. **Start here:** `Models/Pokemon.swift`
   Learn structs, protocols (Codable, Identifiable, Hashable), computed properties, and custom JSON decoding.
2. **Then:** `Models/PokemonType.swift`
   Understand enums with associated values, CaseIterable, and how to attach display data (colors, icons) to enum cases.
3. **MVVM pattern:** `ViewModels/PokemonListViewModel.swift`
   See how @Published, @MainActor, ObservableObject, and Combine work together to manage UI state.

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4. **Networking:** `Services/PokemonAPIService.swift`
   Learn async/await, URLSession, Codable decoding, and typed error handling with a custom error enum.
5. **SwiftUI basics:** `Views/PokemonRowView.swift` and `Views/TypeBadgeView.swift`
   Understand view composition, AsyncImage, @EnvironmentObject, and accessibility.
6. **SwiftUI advanced:** `Views/PokemonListView.swift`
   Study NavigationStack, .searchable, .refreshable, infinite scroll, LazyVGrid, and value-based navigation.
7. **Custom layouts:** `Views/PokemonDetailView.swift`
   Learn the Layout protocol, GeometryReader, and async state handling.
8. **Testing:** `PokedexTutorialTests/`
   See how to write unit tests with XCTest using the AAA pattern with mock services.

7 Command Line Quick Reference

For users comfortable with Terminal:

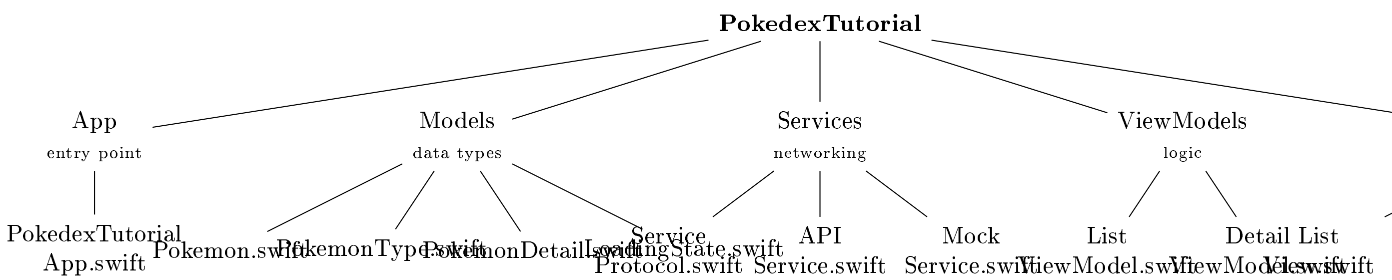
```
# Build the app
xcodebuild -project PokedexTutorial.xcodeproj \
  -scheme PokedexTutorial \
  -destination 'platform=iOS Simulator,name=iPhone 17 Pro' \
  build

# Run all tests (24 tests, ~2 seconds)
xcodebuild -project PokedexTutorial.xcodeproj \
  -scheme PokedexTutorial \
  -destination 'platform=iOS Simulator,name=iPhone 17 Pro' \
  test

# Launch in simulator
xcrun simctl boot "iPhone 17 Pro"
xcrun simctl install "iPhone 17 Pro" \
  ~/Library/Developer/Xcode/DerivedData/PokedexTutorial-*/Build/Products/
      Debug-iphonesimulator/PokedexTutorial.app
xcrun simctl launch "iPhone 17 Pro" com.enigmak9.PokedexTutorial

# Open in Xcode
open PokedexTutorial.xcodeproj
```

8 Project Structure Reference



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Appendix: Simulator Keyboard Shortcuts

Shortcut	**Action**
Cmd + 1	100% scale (actual size)
Cmd + 2	75% scale
Cmd + 3	50% scale
Cmd + 4	33% scale
Cmd + 5	25% scale
Cmd + K	Toggle software keyboard
Cmd + Shift + H	Go to Home screen
Cmd + Shift + H H	App switcher (double-press)
Cmd + Left/Right	Rotate device

This guide was written as part of the Pokédex Tutorial project.
For technical documentation, see `docs/architecture.md` and `docs/flowcharts.md`.
For the quick-start guide, see `docs/run.md`.

Questions? Open an issue in the repository.

Created by **enigmak9** • July 2026

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